that band you can make out the blue sky fading into the blackness of space. This is the transition zone that Simons was the first to enter and Gagarin the first to observe from above. In routine spaceflight, you start at the bottom of the blue, penetrate entirely through it a few minutes after liftoff, and then enter that boundless realm where a simple breath of air is impossible without elaborate life-support systems. Human life depends for its very existence on that blue sky. We are right to consider it tender and sacred.

We see the blue in daytime because sunlight is bouncing off the air around and above us. On a cloudless night, the sky is black because there is no sufficiently intense source of light to be reflected off the air. Somehow, the air preferentially bounces blue light down to us. How?

The visible light from the Sun comes in many colors—violet, blue, green, yellow, orange, red—corresponding to light of different wavelengths. (A wavelength is the distance from crest to crest as the wave travels through air or space.) Violet and blue light waves have the shortest wavelengths; orange and red the longest. What we perceive as color is how our eyes and brains read the wavelengths of light. (We might just as reasonably translate wavelengths of light into, say, heard tones rather than seen colors—but that's not how our senses evolved.)

When all those rainbow colors of the spectrum are mixed together, as in sunlight, they seem almost white. These waves travel together in eight minutes across the intervening 93 million miles (150 million kilometers) of space from the Sun to the Earth. They strike the atmosphere, which is made mostly of nitrogen and oxygen molecules. Some waves are reflected by the air back into space. Some are bounced around before the light reaches the ground and they can be detected by a passing eyeball. (Also, some bounce off clouds or the ground back into space.) This bouncing around of light waves in the atmosphere is called "scattering."

But not all waves are equally well scattered by the molecules of air. Wavelengths that are much longer than the size of the molecules are scattered less; they spill over the molecules, hardly influenced by their presence. Wavelengths that are closer to the size of the